Training Task Document: AI Character Simulation with Emotional Adaptation

Objective

The goal of this task is to design and implement an AI chatbot system that enables users to extract characters from history or fiction books and engage in conversations with them. The chatbot will simulate the character's emotions using Dorner's Psi Theory, dynamically updating psychological parameters and emotional states based on user interaction. Additionally, the chatbot should have long-term memory to retain past interactions and recognize previous users.

Task Overview

Participants are required to:

- Develop a system where users can input text or upload books.
- Implement a process that extracts a list of characters from the book using an LLM and allows the user to choose a character to chat with.
- Simulate character emotions using Dorner's Psi Theory:
    - The LLM should generate psychological parameter values (arousal level, valence level, etc.) and emotional states (sadness, anger, joy, etc.) based on the character's general behavior and user input.
    - These parameters and states should be displayed in the UI and updated in real-time after every user message.
- Implement long-term memory so that if a different user asks about a past user (e.g., "Do you know Birhanu Worku?"), the character remembers and responds accordingly.

Requirements

Functional Requirements:

1. Character Extraction:

- Users can enter text or upload a history/fiction book.
- An LLM processes the text and extracts a structured list of characters.
- Users can select a character to engage in conversation with.

2. UI for Parameter Control & Display:

- A user-friendly UI displaying:
    - List of extracted characters
    - Psychological parameters (Valence Level, Arousal Level, Selection Threshold, Resolution Level, Goal-Directedness, Securing Rate)
    - Emotional states (Anger, Sadness, Joy, etc.)
- Parameters and emotional states should update automatically in real-time during conversation.

3. Adaptive Dialogue:

- Chatbot responses should be influenced by:
    - The selected character's predefined personality traits
    - The emotional state calculated dynamically
    - User interactions affecting the character's emotions over time

4. Long-Term Memory:

- Chatbot should retain conversation history beyond individual sessions.
- The system should be able to recall and respond accurately when a different user asks about a past interaction.

Technical Requirements:

Technology Stack:

- **Language Model & Framework:** LangChain (Python) for conversational AI and memory management.
- **Frontend:** React
- **Backend:** Flask (preferred)
- **Database:** Chromadb

Dorner’s Psi Theory

Dorner’s Psi Theory

Psi Theory Parameters:
1. Valence Level: Measures the spectrum of attraction (appetence) vs. aversion; corresponds to positive vs. negative reinforcement.
2. Arousal Level: Reflects the agent’s readiness for action, similar to the function of the ascending reticular formation in humans.
3. Selection Threshold: Indicates how easily the agent shifts between different intentions or balances multiple goals; reflects the dynamics of motive dominance. A higher selection threshold means the agent shifts less easily.
4. Resolution Level: Describes the agent’s accuracy in perceiving the world, ranging from detailed cognition to rapid perception.
5. Goal-Directedness: Represents the stability of the agent's motives; indicates how strongly the agent prioritizes its goals versus adapting or 'going with the flow.'
6. Securing Rate: Refers to the frequency with which the agent checks its environment; involves reflective and orientation behaviors.

Sample of Five Emotions (Directed Affect Plus Modulation) According to Psi Theory:

1. Anger: Arises when an obstacle (often another agent) clearly prevents the achievement of a relevant goal. Characteristics: negative valence, high arousal, low resolution level, high action-readiness, high selection threshold, and goal redirection to counter the obstacle.

2. Sadness: Occurs when all perceived paths to achieving active, relevant goals are blocked, without a specific obstacle. Characteristics: negative valence, low arousal, decreased action-readiness due to goal inhibition, and an increased demand for affiliation (support-seeking behavior).

3. Pride: Emerges from a perceived high level of self-competence and internal legitimacy, often coinciding with high external legitimacy. Accompanied by a low securing rate.

4. Joy: Characterized by high arousal and a strong reward signal from fulfilling a demand, typically related to achieving meaningful, higher-order goals.

5. Bliss: Defined by low arousal with a high perceived reward signal from fulfilling a demand, generally associated with higher-order cognitive demands, such as aesthetics. Bliss is not primarily linked to physiological demands, which usually involve high arousal. It has a transcendent quality that aligns with serotonergic signaling and creates a state of consciousness that transcends personality dynamics, though personality can influence how bliss is accessed and experienced.